Mumbi Whidby

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Objective

Robotics researcher and hands-on hardware engineer building **tactile sensors** and **manipulation systems** for robots, from sensor fabrication through **Vision-Language-Action-based control**, plus **haptic feedback** for **sensing-degraded environments** such as underwater settings. Seeking a hands-on R&D or robotics engineering role building novel sensing and manipulation systems for real-world robotic platforms.

Education

University of California, Los Angeles — Los Angeles, CA
Ph.D., Mechanical Engineering (Design, Robotics, and Manufacturing) — *Sept. 2022–Present*
M.S., Mechanical Engineering — *March 2024*

Spelman College — Atlanta, GA
B.S., Computer Science, *Magna Cum Laude* — *May 2022*

Technical Skills

Programming: Python, C++, MATLAB
Tools/Frameworks: ROS, ROS2, OpenCV, SolidWorks, Onshape, Arduino, Git
Fabrication: soldering, silicone molding and casting, basic circuit assembly

Research Experience

UCLA Biomechatronics Lab — Graduate Researcher — *Sept. 2022–Present*
Advisor: Prof. Veronica J. Santos

- Designed, fabricated, and characterized a tactile sensor skin for robotic fingertips (EGaIn liquid-metal taxel from University of Washington collaborators), evaluating four configurations across core geometry, overmold thickness, and taxel placement via benchtop and real-world grasping tests teleoperating a LEAP hand with MANUS Metagloves Pro.
- Developing a VLA-based robotic manipulation system with a custom dual-modality tactile sensor to infer and compare physical object properties across sequential grasps.
- Designed and built a wearable haptic feedback prototype to restore tactile discrimination in sensing-degraded environments, such as underwater settings.
- Mentored six undergraduate researchers across sensor fabrication and manipulation research.

Robotic Platform for Diagnostic Flexible Laryngoscopy — Co-Hardware Lead — *Aug. 2025–Present*

- Co-developing an autonomous robotic system for diagnostic flexible laryngoscopy with UCLA Geffen School of Medicine otolaryngologists, validated through cadaver testing; filed an invention disclosure leading to a provisional patent.
- Placed 1st in the UCLA Lowell Milken-Sandler Prize and UCLA Knapp Venture Competition, and 3rd in the Easton Innovation Challenge; completed the NSF I-Corps Regional Program.

Human Fusions Institute, Case Western Reserve University — Research Assistant — *Oct. 2023–Sept. 2025*

- Evaluated glove-based hand-tracking hardware and demonstrated a closed-loop tactile teleoperation system (LEAP hand, tactile sensors, haptic display rings) for an ONR-funded bimanual teleoperation project.

NASA L'SPACE Mission Concept Academy — Researcher — *Jan.–May 2020*

- Designed payload systems and co-authored the Preliminary Design Review for a NASA Lucy Mission concept team.

U.S. Army HBCU/MI Drone Design Competition — Engineering Team Member — *April 2019*

- Integrated flight control hardware and sensors into a bio-based UAV airframe and conducted flight testing as remote pilot.

Industry Experience

Honeywell Enterprise — Launch IT Intern — *May–Aug 2021*

- Investigated robotics integration opportunities (AS/RS, AMRs) across U.S. and U.K. sites; designed AWS-based cloud solutions for robotic platform deployment and monitoring; conducted cost–benefit analyses across fulfillment scenarios.

Publications

Whidby et al., "Design Methodology for a Microfluidic Tactile Sensor Skin for Robot Fingertips." In preparation. ·
Harber et al. (incl. Whidby), "OptiStrain: A Vision- and Microfluidics-based Tactile Sensor." In preparation.

Awards and Honors

Cota-Robles Fellowship · Boeing Scholar · Upsilon Pi Epsilon